

TERESINA, Brazil

WMO: 825790

Lat: 5.060S

Lon: 42.823W

Elev: 219

StdP: 14.58

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	71.4	72.6	52.2	58.5	97.3	55.1	65.0	96.2	12.3	80.4	10.8	81.6	2.0	180	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	22.8	102.1	72.1	100.6	72.7	99.0	73.2	81.0	87.7	80.4	87.5	79.9	87.1	5.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.8	83.3	78.9	151.4	83.1	78.5	149.0	82.8	44.9	87.4	44.2	88.4	43.6	87.5	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 12.0	10.7	9.5	DB	67.5	103.8	1.8	1.9	66.3	105.2	65.2	106.3	64.2	107.3	62.9	108.7	(4)
(5)			WB	63.7	83.9	1.6	1.9	62.5	85.3	61.6	86.4	60.7	87.4	59.5	88.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	84.9	83.6	82.9	82.8	83.0	83.5	83.3	83.8	85.3	87.4	88.4	87.7	86.3	(6)
(7)		DBStd	2.85	2.71	2.27	1.92	1.67	1.49	1.52	1.79	1.83	1.53	1.79	2.38	2.82	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12724	1042	922	1017	991	1039	1000	1046	1096	1123	1192	1132	1124	(10)
(11)		CDD65	7249	577	502	552	541	574	550	581	631	673	727	682	659	(11)
(12)		CDH74	87863	6353	5200	5518	5513	6379	6267	6900	8179	9320	10287	9427	8520	(12)
(13)		CDH80	45001	2926	2267	2304	2268	2735	2864	3446	4546	5490	6156	5431	4568	(13)
(14)	Wind	WSAvg	3.6	3.3	3.0	2.9	2.8	3.1	3.5	3.9	3.9	4.1	4.4	4.2	3.9	(14)
(15)	Precipitation	PrecAvg	54.8	8.1	8.9	12.0	10.6	4.7	0.7	0.5	0.4	0.6	0.9	2.6	4.9	(15)
(16)		PrecMax	101.1	19.9	19.9	24.6	25.6	12.1	3.9	2.8	3.1	3.3	3.6	8.5	16.3	(16)
(17)		PrecMin	27.8	2.4	1.6	0.0	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	(17)
(18)		PrecStd	14.5	4.4	4.0	5.6	6.2	3.6	0.8	0.7	0.8	0.8	1.0	2.2	3.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.8	96.0	95.4	94.4	95.0	95.3	97.4	100.2	102.5	104.0	103.6	101.8	(19)
(20)		MCWB	74.8	77.9	79.3	79.7	78.1	74.9	72.5	72.0	71.7	72.4	71.6	72.4		(20)
(21)		2%	DB	96.2	93.6	93.1	92.9	93.2	93.5	95.4	98.6	100.7	102.2	101.6	99.0	(21)
(22)		MCWB	76.0	78.0	79.2	79.3	78.0	75.1	72.8	71.8	72.1	72.1	72.6	74.2		(22)
(23)		5%	DB	93.5	92.0	91.5	91.4	91.7	92.6	94.6	97.0	99.2	100.6	100.1	97.3	(23)
(24)		MCWB	77.0	77.9	78.7	78.9	77.9	75.2	72.8	72.0	72.5	72.7	73.3	74.8		(24)
(25)		10%	DB	91.6	90.5	89.8	89.8	90.7	91.3	93.0	95.4	98.3	98.9	98.2	95.4	(25)
(26)		MCWB	77.2	77.9	78.4	78.7	77.8	75.3	72.9	72.3	72.6	73.5	74.0	75.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	81.4	81.6	81.9	81.6	79.8	78.7	77.8	77.8	78.8	79.5	79.9	(27)
(28)		MCDWB	88.5	87.4	88.3	87.9	87.5	86.2	85.5	88.2	90.8	89.9	89.7	89.4		(28)
(29)		2%	WB	79.7	80.1	80.6	80.9	80.5	78.9	77.5	76.8	76.9	77.7	78.5	79.0	(29)
(30)		MCDWB	87.9	87.4	88.0	87.6	86.9	85.5	85.2	87.3	89.8	89.7	89.0	88.8		(30)
(31)		5%	WB	79.1	79.6	79.9	80.2	79.8	78.2	76.6	76.0	76.2	77.0	77.7	78.3	(31)
(32)		MCDWB	87.4	87.1	87.3	87.1	86.3	85.2	84.9	86.6	88.7	89.5	88.5	88.1		(32)
(33)		10%	WB	78.4	79.0	79.3	79.7	79.2	77.5	75.9	75.2	75.5	76.3	77.1	77.7	(33)
(34)		MCDWB	86.7	86.7	86.6	86.4	85.8	84.9	84.6	85.8	87.6	88.7	88.0	87.3		(34)
(35)	Mean Daily Temperature Range		MDBR	16.8	16.0	15.7	15.4	15.5	17.3	19.7	22.5	23.2	22.8	20.8	18.8	(35)
(36)		5% DB	MCDBR	19.2	17.5	17.1	16.6	16.6	18.5	21.3	23.5	24.3	24.5	23.6	21.3	(36)
(37)			MCWBR	4.9	5.5	5.7	5.6	4.8	4.8	5.1	5.3	5.0	5.0	4.6	4.4	(37)
(38)			5% WB	MCDBR	16.6	16.1	15.9	15.4	15.3	16.7	18.5	21.5	22.6	21.8	19.4	18.4
(39)		MCWBR		5.8	6.2	6.1	6.0	5.5	5.1	5.2	5.6	4.9	4.9	4.8	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.423	0.410	0.413	0.404	0.399	0.384	0.373	0.383	0.442	0.474	0.469	0.454	(40)	
(41)		taud	2.417	2.466	2.462	2.492	2.488	2.524	2.548	2.482	2.300	2.227	2.245	2.306	(41)	
(42)		Ebn at Noon	290	294	289	284	277	277	282	286	276	272	275	280	(42)	
(43)		Edh at Noon	39	37	37	35	33	31	31	35	43	47	46	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1685	1708	1717	1668	1697	1732	1835	2036	2114	2052	1915	1760	(44)	
(45)		RadStd	102	112	72	74	76	57	60	51	72	62	98	77	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.74	+0.45	-4.36	+1.49	N/A	N/A	N/A	N/A	+230	+230
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page